

MOHAMED OSAMA ZAHARAN

Full-Stack Engineer

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PROFESSIONAL SUMMARY

Full-Stack Engineer with 2+ years building production web applications end-to-end: React/Next.js frontends backed by NestJS and ASP.NET Core APIs, with applied AI (LLM tool-calling agents, voice pipelines, NLP services). Delivering 4+ production systems at RICOH Europe alongside client products across e-commerce, healthcare marketplaces, CRM, and fintech workflows. Strong testing culture (390-test agentic suite), real-time systems (SignalR/WebSocket), Arabic RTL/i18n, and Docker-based delivery in multicultural teams.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), C#, Python, SQL

Frontend: React, Next.js, Redux Toolkit, Zustand, TanStack Query/Table/Form, React Router, React Three Fiber / Three.js

Backend: NestJS, ASP.NET Core, EF Core, MongoDB / Mongoose, SQL Server, PostgreSQL, Prisma ORM, SignalR, WebSockets, Hangfire, REST APIs, CI/CD

AI & Integrations: LangGraph multi-agent pipelines, LLM tool-calling + SSE streaming, HuggingFace Transformers (Llama 3, Qwen 2.5, Whisper, DistilBERT), Google Gemini, Redis

Styling & UI: Tailwind CSS v4, Shadcn / Radix UI, Ant Design, Framer Motion, GSAP, Lenis, RTL + next-intl i18n, WCAG accessibility

Tools & Practices: Git, Docker & docker-compose, Vitest, xUnit, Zod, Swagger/OpenAPI, Stripe, Paymob, Bosta API, MinIO S3, NextAuth/JWT, Agile/Scrum

PROFESSIONAL EXPERIENCE

RICOH Europe — Maadi, Cairo

Nov 2025 – Present

Front-End Developer

- Deliver 4 production frontend systems in parallel for cross-regional European operations under the CORELIA umbrella — real-time tailgating detection, 3D building viewer, invoice management, and NLP tooling — meeting European compliance standards.
- Collaborate daily with multicultural, cross-functional teams in a corporate European environment.

NedSwiss — Remote, Switzerland

Feb 2025 – Nov 2025

Front-End Developer

- Built a multilingual admin/staff CRM dashboard (Next.js 15, TanStack Table/Query, Recharts, react-quill) with role-based navigation, rich data tables, chart analytics, and real-time cache invalidation.
- Developed the company marketing website with advanced scroll animations (GSAP, Lenis), next-intl i18n routing, Zod-validated lead capture, and Turbopack bundle optimization.

GDSC – Shorouk Academy — Shorouk City

Aug 2023 – Jun 2025

Frontend Lead

- Led the frontend track of the Google Developer Student Club, running hands-on workshops in React, Next.js, and TypeScript for 50+ students.
- Mentored developers in React, Next.js, TypeScript, and modern UI architecture patterns.

PROJECTS

Medical System — Gov Healthcare Insurance Portal (Aug 2026)

Next.js 16, React 19, TypeScript, TanStack Query, React Hook Form + Zod, next-intl (AR/EN RTL), Recharts, Shadcn / Radix UI, Tailwind CSS, Biome, Vitest

- Co-built a production government healthcare portal (citizen, doctor, and admin portals, live on Vercel) with a 2-developer team, contributing the majority of commits: the full admin verification workflow across 4 phases (application review queue, card lifecycle management, category configuration, verification workbench).
- Built the doctor clinical workspace (point-of-care desk with patient chart and visit management) and the citizen insurance experience (application, tracking history, card-issuance stepper) on OpenAPI-snapshot API contracts with role-based auth.

Nabdh Platform — Home Nursing Marketplace Backend (Jun 2026 – Present)

NestJS 10, TypeScript, MongoDB 7 (2dsphere geospatial), WebSocket, MinIO S3, Docker

- Engineered the modular-monolith backend for an Egyptian home-nursing marketplace: auth, booking, payment, chat, and notification modules serving patient and nurse Flutter apps as well as a React admin panel.
- Geospatial matching with MongoDB 2dsphere indexes, real-time WebSocket chat, and S3-compatible media storage via MinIO.

agentic-shop — Agentic AI Shopping System (Aug 2026)

Python, FastAPI, LangGraph, Pydantic v2, Next.js 16, React 19, SSE

- Built a multi-agent shopping pipeline (LangGraph) that turns a one-sentence goal into ranked products, a conversational answer, and a validated UI plan rendered client-side over SSE streaming.
- Deterministic product scorer with Pydantic v2 contracts; fully offline mock mode with 390 passing tests (224 backend + 166 frontend); real-LLM swap-in ready.

OZ Uniform Store — Full-Stack Client Product (Jul 2026 – Aug 2026)

Next.js 16, React 19, Tailwind CSS v4, TanStack Form + Zod 4, .NET 10 Web API, EF Core 10, SQL Server, Hangfire

- Arabic-first RTL e-commerce storefront for a school-uniform retailer: school/grade catalog, cart, cash-on-delivery checkout, Bosta courier shipping, and per-order token tracking URLs.
- Built the .NET 10 backend: JWT auth, Hangfire background jobs (auto-cancel orders, email), FluentValidation, Bosta shipping integration, and audit logging; deployed to production.

Gym Management API (Aug 2026)

ASP.NET Core 8, EF Core, SQL Server, xUnit, Swagger/OpenAPI

- Single-tenant gym subscription API with cryptographic QR check-in: atomic session deduction under concurrency (SQL aplocks + conditional updates), preventing negative balances.
- Time-based and per-sale session-credit subscriptions, RFC 7807 errors with machine-readable reason codes, structured logging, health checks, and an xUnit test suite.

RICOH Production Work — CORELIA (Nov 2025 – Present)

- **Tailgating Detection System:** real-time anomaly dashboard with live MJPEG streaming, license-plate OCR, role-based access, and audit trails (React 18, Redux Toolkit, TanStack Query, Shadcn UI).
- **3D Building & Space Viewer:** multi-mode 3D viewer (Dollhouse / Panorama / Split) with room-aware navigation, animated overlays, and progressive panorama loading (React 19, React Three Fiber, GSAP, Framer Motion).
- **AGL Invoice Management:** full invoice lifecycle with React Hook Form + Zod integrated with CV team and AI Team for the full pipeline.
- **Qalam Vision:** Arabic/multilingual NLP keyword-extraction interface integrating Label Studio.

Wasla CRM — Multi-Tenant SaaS (Graduation Project) (Nov 2025 – Jun 2026)

Next.js 15, ASP.NET Core 8, FastAPI, SignalR, Hangfire, Stripe, Paymob, HuggingFace

- Full CRM platform spanning frontend, .NET 8 backend, and two AI microservices: multi-tenant JWT auth with per-user permissions, SignalR real-time notifications, Hangfire scheduled jobs
- [AI layer](#): Llama-3.3/Qwen tool-calling backend, and **toxicity moderation** (DistilBERT / DeHateBERT) with sub-100ms latency on high-confidence predictions.

CyrusLearn — Interactive Learning Platform (WE Innovative Hackathon) (Mar 2026 – Apr 2026)

React 19, Vite, TypeScript, React Three Fiber, Redux Toolkit, Shadcn UI

- Feature-driven e-learning platform with 3D interactive labs, structured learning paths, a quiz engine, collaborative rooms, Google OAuth, subscription flow, and an admin dashboard.

AI-Powered Translation Platform — Freelance (2024)

Next.js, TypeScript, Tailwind CSS, next-intl, Paymob

- Multilingual translation product (Arabic/English/French) with end-to-end Paymob billing, a daily-quota credit system, admin transaction monitoring, and a blog module.

EDUCATION

BSc in Computer Science — Shorouk Academy, Cairo

GPA: 3.8/4.0